

UNIT - V

INSTRUCTIONAL MEDIA

In this unit we will see the types of different teaching aids to explain the content in the classroom.

5.1 Characteristics and advantage of using Teaching Learning Materials

Teaching aids are necessary for effective learning. There are many aids like audio, visual and audio-visual such as Cassette, Flash Cards and Television. Senses are the gate ways of knowledge. Useful teaching and learning starts from the experiences of senses. Teachers has to use more number of teaching aids to make teaching effective.

Characteristics of Good Teaching aids

After characteristics of good teaching aids as follows

1. Teaching aids are large enough to be seen by the Students for whom they are used.
2. Teaching aids are meaningful and they always stand to serve a useful purpose.
3. Teaching aids are upto the mark end up to the date in every respect.
4. Teaching aids are simple, cheap and may be improvised
5. Teaching aids are accurate and realistic
6. Teaching aids are according to the mental level of students.

7. Teaching aids purpose may be informative but it is not just entertainment.
8. Teaching aids helps in realization of stipulated learning objects.
9. Teaching aids can be used in many lessons and at different class levels.
10. Teaching aids are useful for supplementing the teaching process but they cannot replace the teacher.

Advantages of Teaching aids

1. Supplement in verbal instructions.
2. Teaching aids makes learning permanent.
3. Teaching aids are helpful in attracting attention of the students.
4. Teaching aids serves time & energy.
5. Teaching aids encourages the healthy classroom interaction.
6. Teaching aids helps the teacher to create situations for teaching and beginners.
7. Teaching aids are helpful in creating +ve environment for discipline.
8. Teaching aids are helpful in meeting individual differences.
9. Students can learn better, through different teaching aids.

10. Teaching aids enable the children to retain the subject matter for a longer time.
11. Teaching aids makes the abstract ideas concrete and helps in making learning more effective.
12. Teaching aids help in the development of various skills among the students.
13. Teaching aids can facilities the proper understanding to the student which discourage the act of cramming.
14. Teaching aids make classroom live and active.
15. Teaching aids provide direct experience to the student.

Conclusion

Teaching with the help of achieve learning objectives. It helps you to think critically and develop skills such as problem-solving, decision making and better understanding. Visual aids increases the classroom learning by 400 percent. The following strategies may be implemented to grab the attention of students in the classroom.

1. Deliver lecture with the help maps, images, charts, etc.
2. Use videos and skill images in your presentation.
3. Demonstrate the concepts with the help of plays, drarnas outdoor and indoor activities.
4. Draw pictures with colour to explain new concepts.
5. Avoid using large texts for teaching.

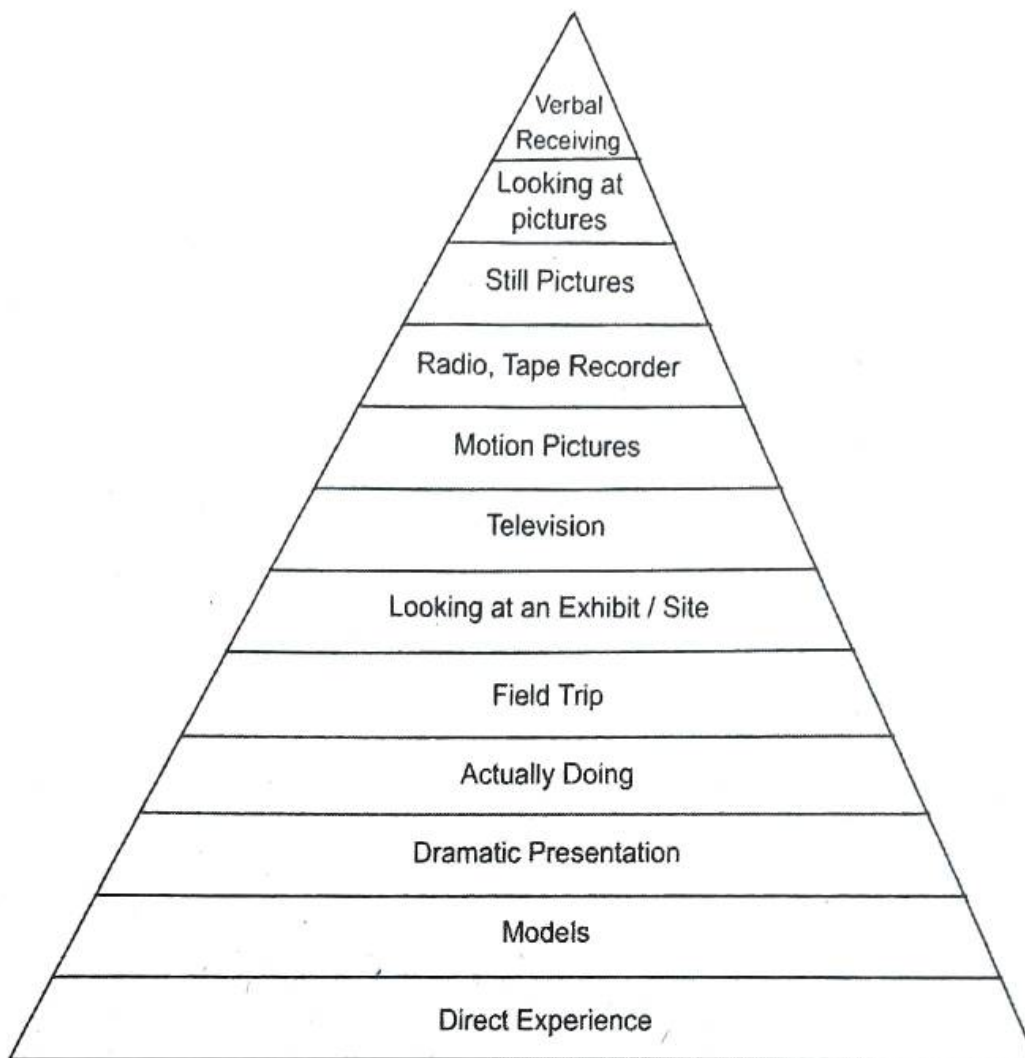
5.2 Psychological Bases of Hardware and Software Technologies

Psychological helps both teacher and learner in accordance with the aim and objectives of education. Hardware and Software Technologies makes educational process interesting, purposeful and effective.

5.2.1 Edger Dale's Cone of Experience

In schools students get knowledge with the help of sensory organs. If we use our sensory organs in learning, it will give constant knowledge. Thou we use our all organs, we use our eyes (83%) and ears (11%) to get knowledge. To explain the subject matter and to make students to involve in teaching learning activities, teaching materials are more helpful.

American educationalist Edger Dale says, generally we can divide the experiences that we get into three sources. First the experiences we gets through the actions. Secondly the experiences we gets through observation and thirdly the experiences we get through abstract concepts. History is a subject of abstract concepts. If we teach it as it is, then students will not understand properly and the learning will not happen as we expect. Hence learning is uninteresting. In order to explain abstract concepts, we have to use audio-visual aids to teach History and to create interest to the students. Edger Dale logically divide all audio visual aids into subdivisions and put into order in the shape of a cone.



Edger Dale's Cone of Experience

Here I recall one Chinese proverb.

I hear, I forget

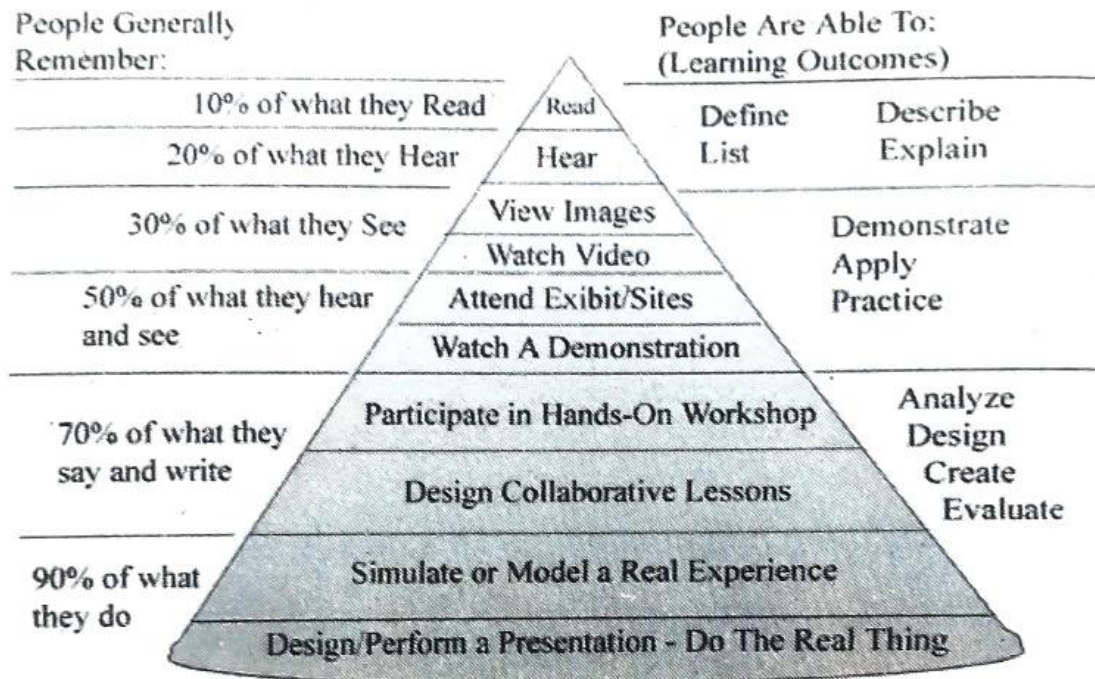
I see, I remember

I do, I understand

While teaching, teacher has to provide direct experiences. If it is not possible then teacher has to give the experiences in order shown by Edger Dale in his cone of experience. Teacher has to give lecture method of teaching only at the end if all the other experiences are not available.

Using Edger Dale's cone of experience in teaching of History

Generally students recall their knowledge in the following manner.



Dale's Cone of Experience

Edgar Dale's mastery cone goes from direct experience (bottom) to the ideological stage gradually (upwards).

Teaching is a process that connects students and the subject matter. It is the duty of the teacher to know the student and his environment and to formulate teaching activities accordingly. The teacher, the student, the teacher's syllabus and the teaching method lead to the best way of teaching. Teaching should make learning enjoyable. The teacher should know the interest of the students and their mood and carry out the teaching process accordingly. To carry out teaching without students understanding what the teacher is doing is to lose its purpose.

Course knowledge should be motivated for students to self-discover. This is what true learning is all about. Teachers need to support this. Teachers need to minimize over-explaining and direct feedback when talking too much.

Naturally students will be curious to know and be talented with new force. It is better to teach knowledge to students than to teach it to them.

Albert Einstein they "I do not teach my students. I will create opportunities for them to learn" he said many years ago.

Edgar Tale's mastery column states that teaching should focus on students' senses, and that learning occurs more when more senses are used. It is important to teach students accordingly with appropriate models. Here we see the statement made by William Arthur.

"The ordinary teacher says,

A good teacher explains,

The best teacher does it.

A great teacher robs the mind"

Edgar Tale's mastery cone also teaches how to carry out teaching to retain students' memory. Students learn a lot by doing "learning". The use of the necessary tools, equipment, and models will maximize teaching and learning ability. Much of the learning takes place at the base of the workshop cone.

- Learning-Teaching activities should be carried out on the basis of experience.

- As far as teaching in History is concerned, all students should set up direct experiences and engage in teaching. If not, the teacher can impart knowledge to the students by doing it.
- Use various visual question tools to enhance the learning of individual students.
- The teacher should plan and undertake field trips according to the course knowledge of the students.
- Make arrangements to take students to science fairs and planetariums.
- The lesson should be conducted with visual quiz tools available in the school or with tools and concepts prepared by the teacher himself according to the lesson concept.
- Avoid teaching History without teaching tools or visual questioning tools.
- Teaching should be done with real life contexts.
- Avoid the mentality of having to take auxiliary tools to class and select and use in the classroom the tools that are located in such a way as to explain the lesson concept.

5.2.2 Multi-sensory teaching

Teaching through multiple senses is an important teaching method. Learning can be best practiced through multiple sensory learning. Learning involves the simultaneous use of visual, auditory, and kinetic-tactile senses. For example the language we see, the language we hear and the language we perceive are constantly making connections between the paths of study and learning.

Teaching through multiple senses is an important aspect of teaching for dyslexia students. It is used by medically trained teachers. Effective teaching for students with dyslexia is transparent, direct, and holistic. It focuses on the structure of language. It pays these students the basics of language, the sounds. It allows these students to slowly and completely teach the basic elements of the language, the sounds and the letters that refer to them, how to combine and separate them. They need to be trained a lot in working together with the sense of hands, eyes, ears and voices they are writing about.

Many are important principles of emotional teaching

- Teaching is used to improve memory and learning through all the learning pathways in the brain (visual, auditory, tactile) simultaneously.
- The teaching sequence should start with the easiest and most basic concepts and progress systematically to the most difficult subject. Each concept should be based on what has already been learned.
- Students - The teacher should teach all concepts directly with continuous communication.
- The teacher must be flexible or proficient in personal teaching. The student should progress gradually and master.

Many sensory learning techniques

Multiple sensory learning includes visual, auditory, muscular and tactile and other senses such as smell and taste. Multiple sense learning differs from the

general learning style. Reports suggest that the human brain has evolved to process multiple sensory signals. Some studies suggest that the benefits of learning are greatest if multiple emotions are involved in learning at the same time. Studies show that multi-sensory learning is best for people over the age of eight.

Most teaching techniques take place through sight or hearing. In reading information, the child's vision is used to read information from text, pictures or board. Many sense teaching is not just about studying and listening. Rather it tries to use all the senses. Each lesson does not use the five senses (taste, smell, touch, sight, hearing and drive). But the vast majority of many emotional subjects involve the child learning in more than one way.

Suppose, for example, that a class reads about apples. Children will have the opportunity to see, touch, smell and taste the apple. Then they can even tell the number of seeds in the apple cut in half. This is the teaching of many senses. Here information is revealed through touch, sight, hearing and taste.

Learning Styles

1. Visual teaching

In this way students can learn by watching the show.

- Viewing the map
- Viewing image
- Watching a movie

2. Aural teaching

In this way students can learn by listening.

- Listening to the lecture
- Listening to the discussion
- Listening to the story

3. Tactile teaching

- This method allows students to learn by touch.
- Touching and manipulating an object
- Conduct a live test

4. Teaching dynamics

This method students learn while moving.

- Through the game
- On overseas trips
- Conduct interactive tests

5. Oral teaching

This time the students are able to know the information by talking about it.

- Divide students into discussion groups
- Encourage students to verbally rehearse their understanding of information
- Telling students to think aloud

6. Direct experiential teaching

Allows students to learn through experience.

- Conducting experiments
- Making field trips.
- Participate in a vocational training program

Multi-sense teaching is an effective strategy

One of the most important challenges facing educators is how to teach students who are struggling with learning. You can prepare as many students as possible in teaching if you incorporate certain strategies in your classroom while students learn at different speeds. Many sensory teaching activities are based on whole brain learning. It is believed to be the best way to teach ideas by covering many parts of the brain. Multi-emotional teaching can help students develop or strengthen their abilities when applied to all students, not just dyslexia, to help them develop strong writing skills by adding hearing and visual elements (online) to reading tasks.

Benefits of multiple sensory learning

1. If a student learns using more than one sense then that information will stay with him for a long time.
2. This method is very beneficial for students with dyslexia.

5.3 Hardware Instructional Aids :

Motion pictures, tape recorders, television, teaching machines, Computers are called educational hardware. Hardware approach mechanises the process of teaching so that teachers would be able to deal with more students with less expenditure in educating them.

5.3.1 Motion Pictures

A series of picture projected on a screen in rapid succession with objects shown in successive positions slightly changed so as to produce the optical effect of a continuous picture in which the object move or cinema.

In 1888 in Newyork, city, the great inventor Thomas Edison and his British assistant William Dickson create a device that could record moving pictures, a primitive motion pictures camera.

A film, also called a movie, motion picture or moving picture is a work of visual art used to stimulate experiences that communicate ideas, stories, perceptions, feelings, beauty etc. These images are generally accompanied by sound and more rarely other sensory stimulations. The word Cinema short for Cinematography is often used to refer to film making and film industry.

The moving images of a film are created by Photographing actual scenes with a motion picture camera, by photographing drawings or miniature models using traditional animation techniques and computer animation or a combination of other visual effects.

Motion pictures as an aid to Education

The idea of using motion pictures in education is as old as the motion picture itself. From the day of this invention, 'educational' possibilities of the motion picture have been recognized. Edison himself generally called the "father of mother picture" believed from the beginning that the Chief contribution of the

motion picture would be to education. Most of the early attempts towards the protection of educational films incurred losses. Educators are also not satisfy with the results. Educators began to understand that application of motion pictures to education was more than just showing a film.

Uses of Motion pictures in Education

The motion pictures is perhaps one of the most powerful agencies in society. The most important function, as for as education is concerned is insight. Motion pictures may present a Veritable and definite portrayal of men's habitat. They may offer a knowledge of a relationship of men. Mc Clusky states that "the power of motion picture to develop behaviour and attitudes in keeping with educational proposes would, if it were properly used, not only step up the intelligence of our citizenry both with respect to moral, social, economic and political issues in this country but also with respect to world problems. In 1936 more than ten thousand motion pictures silent and sound were used in schools.

5.3.2 Computer

This is the computer age. The use of computers is high in all fields. The use of computer by children going to school today is also commendable. Computer education is a compulsory subject in many schools. Let us first know what a computer is.

A computer is a tool that organizes and stores a given amount of information. Thus details can be read, written and searched. And stores a large amount of

information within itself. It is an automatic machine that operates according to the orders given. Once it has given the details and commands in its memory it will start its work and deliver the results quickly and accurately. They do the calculations that a man does on a monthly basis in a matter of seconds. The speed of the computer is microseconds (10^{-6} s), nanoseconds (10^{-9} s), pico seconds (10^{-12} s). High-capacity computers can complete 3 or 4 million simple calculations without a single error in a second.

Computer are of various types and they can be categories two ways on the basis of size and data handling capabilities on the basis of size, there are five types of computers.

1. Super Computer

They are the biggest and fastest computers in data handling. It can calculate upto ten trillion individual calculations per second. It is used in scientific research to analyse data.

2. Main frame Computer

Main frame computers are designed in such a way that it can support hundreds of thousands of users at the same time. It supports multiple programs simultaneously. It is used for big organisations like banking, telecom sector etc.

3. Mini Computer

Minicomputer is a medium size multi-processing computers. It is used in places like Institute or

departments for different work like billing, accounting etc.

4. Workstation

Workstation Computer is designed for technical or scientific application. It is expensive. It is used to handle animation, data analysis, audio and video creation and editing.

5. Personal Computer (PC)

It is also known as microcomputers. It is basically general purpose. Computer and designed for personal use. It consists of a microprocessor as a Central Processing Unit (CPU), memory, input unit and output unit. This type of computers are Laptops and desktop computers.

On the basis of data handling capacities there are three types of Computers. They are

1. Analogue Computer
2. Digital Computer
3. Hybrid Computer

The use of computer in education

1. The computer is an important accessory for learning. Computer helps you to run a lesson efficiently and enthusiastically through Power Bond.
2. The computer is used to explain the physical formulas with appropriate examples to the average and backward students.

3. The computer is used to draw graphs of semantic equations.
4. The system instantly indicates the binary and cube of the largest numbers.
5. Computer-assisted learning (CAI) computer-assisted self-learning.
6. Computer helps to understand the spontaneous learning of direct learning.
7. The computer is used to draw maps and analog maps in colors.
8. The computer is used by the student to store his details and take the details he needs.

5.3.3 Projector

A projector is an optical device. It creates an image by illuminating an image or moving image on a surface, usually through a projection screen or a small transparent lens on the wall. But the new way projectors are screened directly using laser equipment. A virtual net retina display or retina projector is a projector that displays an image directly on the retina instead of the external screen.

The most common projectors used today are called video projectors. Video projectors are an alternative to previous types of projectors such as slide projectors and head props. Older analog projectors are still used in a few places today. Newer types of projectors screen images using portable lasers or tunes. Technology has grown and today it is possible to drop movies on the

walls of the house via mobile as well. Films are now mostly screened in theatres by digital cinema and video projectors.

Types of projectors

Projectors are commonly used in classrooms, corporate meetings, places of worship, concerts, theatres and home theatres. Depending on the situation you can decide what kind of projectors are needed. Projectors can be divided into three types based on their characteristics. They are

Cathode ray tube (CRT) dropped

Cathode ray tube is a device that uses beams of electrons to create an image on a screen.

Liquid crystal display drop (LCD)

It is a video projector that displays video, pictures or computer data on the screen.

Digital light processing (DLP)

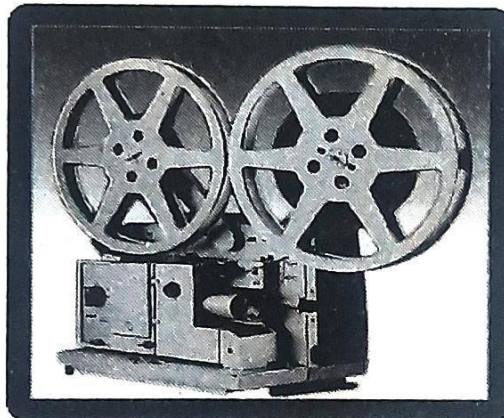
Digital light processing technology captures images by reflecting light from microscopic mirror panels called digital microscopic devices. It works with the best technology.

Projectors for the classroom

Classroom projectors require a projector with a capacity of at least 2500 lumens (brightness). Rooms with high light and rooms with large windows require projectors with a light capacity of 3500 lumens. The more lumens there are in the projector, the brighter the

image will appear. Brightness is very important for two main reasons. If you want to plan in a bright environment you need a bright projector to offset the ambient light. If you want to see the picture on the big screen you need a bright projector.

1. Film Projector



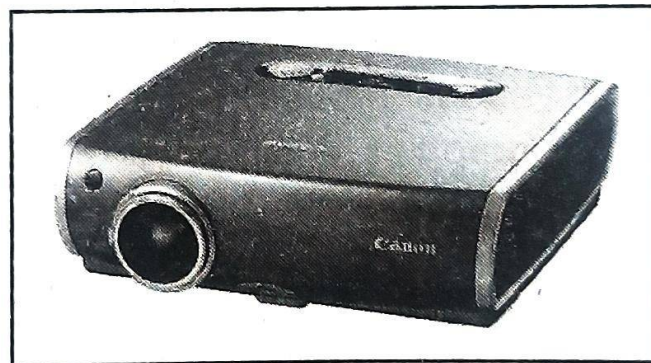
16mm educational films on various historical topics are released by various groups. Films are made on topics such as biographies of historical geniuses and technology. Copies of these films can be purchased or rented at schools. Images can be black and white or in color. Lessons related to the course will be recorded on film. Movies are a great tool for educating most people at once.

Method of use:

1. **Watching the film in advance:** After the picture is brought to the school, the teacher looks at it and can use the picture for which class? Which history subject fits the title of the film? Need to know that. The teacher also needs to know how to engage students in a series of work.

2. **Preparing the student for filming:** The student should be informed of the summary of the film to be seen.
3. **Image display:** Darken the room and adjust the image drop to make sure the image that falls on the screen is clear. If necessary, you can repeat the whole picture or just a few parts.
4. **Continuing Program:** Engage the student in a discussion about what they saw in the film after the film was completed. The teacher asked some questions and asked if the information the student knew was correct. Let's test as.

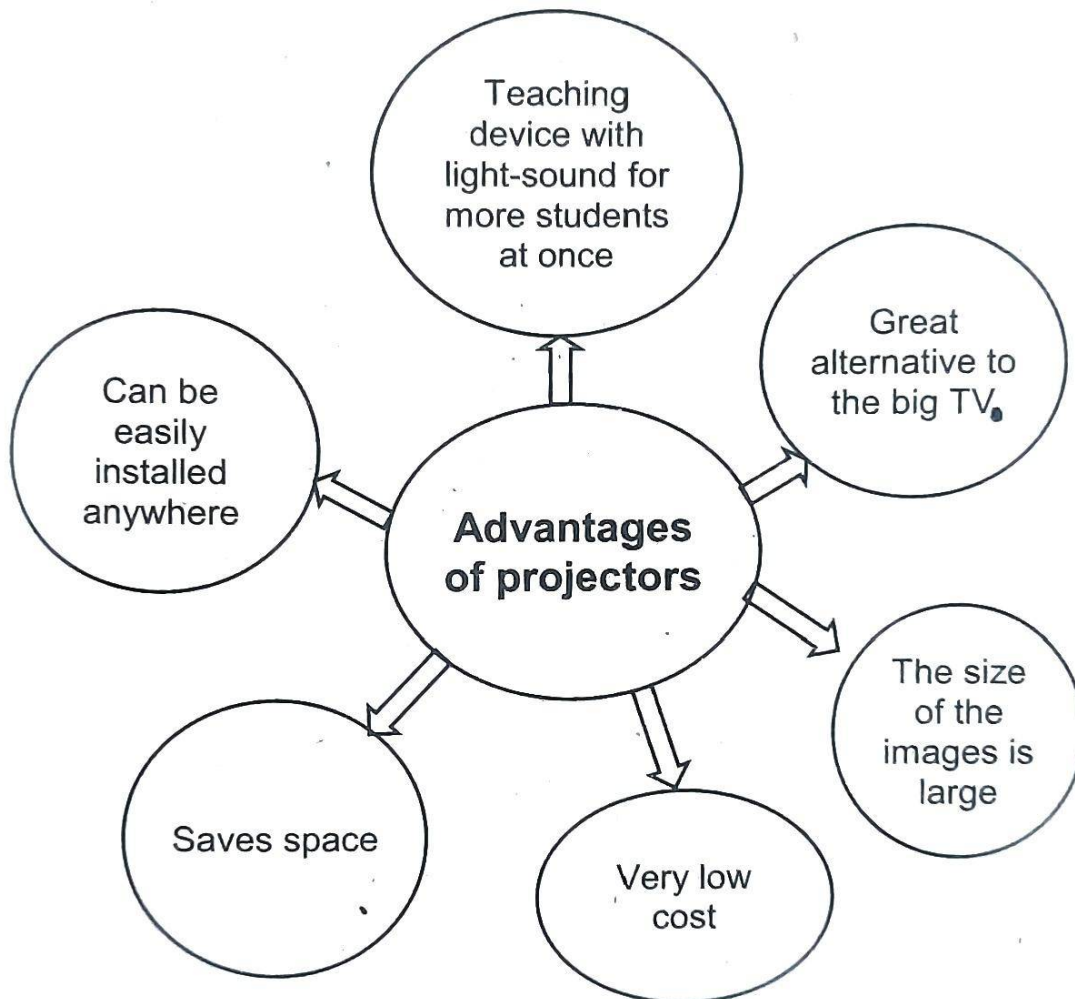
2. Liquid Crystal Display



It is an innovative device with a computer-assisted sound-and-light system. With the help of computer-assisted recording and recording of recorded messages and photos on a CD-ROM 16mm screen, the name LCD can be used to look up a message / image while watching it. It can be used as an image drop over the head and as a motion drop. CDs for history lessons are available at simple prices. You can buy it and show it to students. The teacher can stop the fall where needed and give explanations. Dune is used to stimulate

students' interest and to conduct lessons in an attractive manner.

Advantages of the projector



Disadvantages of the projector

- Dark room is required to use the projector.
- The projector needs maintenance at regular intervals.
- Early installation cost can sometimes increase.
- Most projectors require a separate audio system

5.3.4 Tablet

The tablet is a slightly larger device than a handset. It works as a mobile computer / laptop in many ways. Today tablets are becoming popular due to their simplicity of portability and functionality. We use tablets to browse the Internet, check email, send, play games, download and read books, watch videos, organize content, and more. And these are between the size of a smart phone and a laptop. Tablets vary depending on the keyboard and device displayed on the screen.

Size and capacity of tablets

Tablets are smaller and lighter than laptops. The screen of the laptop is 12 to 15 inches. The screen size of the tablets ranges from 8.5 inches to 10 inches. The weight of the tablet will be less and it will be thinner. Due to this the users consider the tablet to be very convenient for transportation and easy to carry. Laptops offer more features than tablets. Laptops have a storage capacity of 250 lbs. But the tablets have a storage capacity of 16x or 32x. Tablets can run thousands of applications. Tablets are easier to operate than laptops. And tablets are less expensive than laptops.

Advantages of tablets

Educational technology has evolved from learning on the blackboard to learning on interactive white boards. Today education is expanding through tablets, desktops, laptops. There are several benefits to choosing tablets for electrification.

1. Students finish homework quickly using the internet on their tablets.
2. Not every student learns the same way. Students with disabilities find it difficult to learn information like their peers. Tablets help students overcome this difficulty by providing different forms for teaching a lesson.
3. Switching to tablets with electronic textbooks instead of traditional books can save money and there is no need to carry books in the bag.
4. Tablets are moving towards the concept of paperless office. Greatly reduce the use of paper in school. This is a great way to save the environment.
5. Do things that can be done on a smart phone can be done on tablets. The large screen of tablets allows students to function with engagement.
6. Tablets are very easy to carry to the classroom. You can turn on the tablet in a few seconds by pressing a button.
7. The teacher can take notes on important ideas through the tablet while giving a lecture in class.
8. Tablets are used for classroom collective teaching. How to complete a project as a group You can complete the project with the help of tablets.
9. Tablets provide an enhanced experience for distance education.

10. Tablets help students take back and redirect courses they have already taken note of.

5.4 Software Instructional Aids :

Educational software is a term used for any computer software which is made for an educational purpose. Educational software ranges from language learning software to classroom management software. The purpose of all this software is to make education more effective and efficient.

5.4.1 Geotag

Geotag or Geotagging is the process of adding metadata that contains geographical information about a location to a digital map. The data usually consists of latitude and longitude coordinates, but may also include a timestamp, as well as links to additional information. Geotag metadata can be added manually or programmatically.

In Google maps and similar GPS services, geotagging may also be referred to as dropping a pin. Pins can be tagged with contextual information to share information about a specific physical location. Popular types of contextual information includes Photos, Videos, Website URLs (uniform resource locator) and QR Code.

Geotag Pin

The red icon above is called a pin and its use on a digital map indicates that a location has been geotagged. End users can view a pin's additional

metadata by hovering their finger or mouse over the icon.

Location identification has become a fundamental feature of many social media sites and can be a useful tool when added to business applications and medico legal applications. For example some social networking sites and services give out their location of their users, which allows users to know exactly where their friends are as they are logged on to that website. Since devices and tracking are now global, that social media tracking becomes a more powerful and valuable tool for users.

How does Geotagging work

Methodologies for geotagging vary quite a bit, some of them work with geospatial location and 'toponym' information only in an archived context, where others stream content continually or aggregate content for machine learning modules.

Regardless of how they are set up and what they do, geotagging systems will represent the same fundamental idea; that digital tracking of location provides value added services in a given application context.

Importance of Geotags

Geotagging provides helpful insight into consumer activity. Geotags can be used to learn about historical places. Geotags also reveal where individuals are when engaging on the Internet or where they are moving with their mobile device. Within GeoDay, Ciller provides

insight into call traffic that can be used for planning and analysis in the commercial and public sector.

Advantages of Geotags

It gives us the route from where we are to a particular destination, its distance and the time it takes to get there, all in a matter of seconds.

Security experts urge everyday users to be wary of adding too much location data to their profiles or other entries, as criminals could use that data to conduct cyber-attacks or track their physical whereabouts. On the other hand, geotagging is a big boom for law enforcement. In general, all of that geotagging is used to better understand suspected criminal activity.

5.4.2 Charts

History concepts such as history formulas, historical figures, comparisons and differences can be written in large letters and explained to the students during the lesson. Students' memory will be enhanced if they write history pictures, events and ideas and display them in the classroom so that the student can see them every day. The drawing card serves as an inexpensive teaching tool to keep students interested. It saves the teacher time and effort. If the concept pictures are made in attractive different colors it will attract the students. Ideas should not be too small or too large. Must be appropriate for learning.

Ideas for teaching are used for the following purposes.

- To motivate students

- Explain ideas through maps
- Take the main idea of the lesson to the students
- The theme is easy to understand
- To attract the attention of the student
- Used to describe the relationship between history concepts.
- Used to develop scientific thinking.
- Concepts provide a live experience.

The use of concepts reduces teacher time and effort. Ideas are used to attract the attention of students. Typically the size of a concept image is 51cm by 64cm. It is a two dimensional display accessory. The teacher should ask himself these questions before using the concepts.

- Is the subject matter of the concept relevant to the subject matter?
- Are conceptual images appropriate for teaching purposes?
- Is the concept image information accurate? Are the facts clearly stated?
- Will my students understand this visual image?
- Are the concepts attractive and used to motivate students to learn?

Types of Charts

- Image charts
- Term charts

- Map Charts
- Illustrated charts
- Bar Charts
- Round Charts
- Series charts
- Gallery charts

5.4.3 Maps

Drawings have been hanging on classroom walls for generations. Maps are the flat shape of the globe. Maps ~~represent~~ the real world on a very small scale. Where are you? What countries are around you. Where do you want to go. What is the cost? Maps help to get information like which way to go and how fast to go. There are many types of maps. You can tell by the type you use. There are maps on the sides of the paper or on a device with GPS technology. They help you get around in your daily life.

Types of maps

1. Political map

The political map shows the state and national boundaries of a place. There are no landscape features on the political map.

2. Map in nature

A physical map shows the physical features of a country, such as rivers, mountains, forests, and water bodies. Features are usually displayed in different colors. Rivers and lakes are blue, and low-

lying areas are dark green, light green as the height increases, and finally orange. The mountains are shown in brown.

3. Topographic map

Topographic maps are similar to physical maps. It shows the physical features of an area. Spatial maps, however, show elevation and landscape changes, with the help of elevation lines and colors.

4. Climate map

Climate map shows information about the climate of different regions. For example, it shows areas with high rainfall, high snowfall or dry weather. It uses colors to depict areas with different climates.

5. Economic or resource map

Economic or resource maps show the resources in the area and the economic activity that exists there. Shows the types of crops grown there and the minerals found in the area. Symbols are used to depict fertility.

6. Road map

Road map is the most widely used map. It shows different roads, highways or railroads. Road maps are usually generated individually by city. There are road maps for an entire country.

Important drawing skills that students need to know

Students need these important skills as they learn environmental science and Globe. Teaching drawing

skills is an important record of every student's development and education. Students develop spatial reasoning skills by learning how to read maps. As a student develops spatial skills they acquire skills such as understanding symbols, images, objects and their relationships, visualizing them, or understanding their relationship and distance. This can improve students' research and writing skills.

Creating maps with PowerPoint

Internet connection for students in the classroom, view maps through Windows 10 and Microsoft Office 2019 or 365. You can select any country in the world and highlight those cities. We can click and enlarge the images to suit our needs. You can only zoom in on that location by selecting a specific location on the map.

Teaching with a map

Teaching using drawings in the classroom stimulates students' interest. Encourages research and is used to solve problems. Maps can be used to explore multiple topics. Maps can combine visual, learning, spatial thinking and quantitative skills. When using maps the concepts can easily visualize the whole situation.

Benefits of using maps

1. Maps are very simple to use.
2. Easy to carry from one place to another.
3. Maps accurately show all the location of the earth or a specific location.

4. It is possible to know the physical structure of a country and its cultural characteristics through maps.

Disadvantages of using maps

1. It is not accurate to show the three dimensions of the earth in the flat part of the map.
2. Maps without accurate measurements give a bad impression of distance.

5.4.4 Globes

A globe is a three-dimensional sphere used to represent the Earth as a whole. It helps in searching different countries and seas. The distribution of water and land on the Earth's surface is explained by the globe. It shows the exact shape, location and size of the Earth's continents and oceans. So the globe is useful to us. The word globe comes from the Latin word *globus*. The circumference of the earth is about 40 million meters. Many spheres are made with a circumference of one meter.

Globes are important because they help us in History and show the location of distant places unlike a map, globes are an exact representation or model of the earth. Without globes no one would know what the earth would really look like, and there would be no maps because they are based on globes.

The importance of the globe in the classroom should not be underestimated. History and maps are important learning tools in countless subjects such as social studies, History, and science. A globe helps students

understand where they are and where the rest of the world is located, and learn the unique shape of the Earth. Interactive globes have become a staple in many classrooms. These interactive globes can electronically identify certain regions and provide a relative understanding of multiple locations. The interactive world globe combines technology to answer questions and provide information, providing a unique experience for teachers and students alike.

Globes help in understanding and problem solving

Students can develop their reasoning and problem solving skills by studying the maps on the globe. The right type of map will help students keep track of distances between locations and locations on Earth. Students can use globes to estimate the routes and travel destinations of past sailors. This will have a great effect on building self-sufficiency and confidence when creating solutions. New York City has the world's largest globe at 120 feet in diameter.

Globes help students understand History

History is the first subject of students. So you can feel the importance of globes in the classroom. We can use globes for any class. Students will be able to learn about different geographies of places and countries, including water levels, mountain ranges, natural resources and the effect of climate on different places. In a History class, the teacher conducts the lesson with the help of globes.

Linking Culture to Globes

A globalist can conceptualize the different locations of students and learners. By developing a sense of commonality among students, the culture of different countries can be bridged through wars, computer dominance, politics.

Uses of planets

1. A globe is a model of the Earth. It is used to see sea route, air route etc. to find location.
2. Used to know the location of various countries. It shows latitude and longitude lines.
3. Globes help students understand history and History subjects.

5.4.5 Cartoons

Cartoon is a type of expression or message. It is one of the many genres of art with a sense of humor. It is full of satirical political maps and animated films. These images may be unreal or half real. The cartoon first appeared in Punch magazine in 1843. Today children and adults alike love to read caricatures or stories in magazines and newspapers. Animated films have been around since the early 20th century.

Political cartoons appear in almost every newspaper today. Cartoons are used to visualize political events. TV shows and short films are made for cartoonish children. Kids love cartoons and their messages. Notable cartoonists include Scott Adams,

Sarah Anderson, Edgar Henry Banker, Steve Bell, and Virgil Franklin Bath.

If the subject is made into a caricature and taught to the students, the students will learn it willingly. If you take the lessons in a continuous animation mode (in animated image form) and show them to the students, they will happily learn it. Cartoons, especially for school children, are animated and the students will willingly accept the lesson ideas.

Benefits of teaching using cartoon and animation

1. Understand the subject matter through humorous thinking to the imagination of the students.
2. Educational animation software is very good for teaching science and history. Changes that can occur under the microscope can be easily shown through animation. Students can see ideas through animation that are invisible to the naked eye.
3. Animation software is used to enhance the creative ability of students.

5.4.6 Posters

Posters make the message appealing to students. Advertising boards are used to deliver messages, attractively. Posters are used to convey messages and remind. Posters are usually associated with school ceremonies and meetings. The cost per poster to advertise the news is low. The following five ideas should be kept in mind when creating a billboard.

Key properties of the poster

Purpose - To give clear color that is understandable

Saying method - simply, in capital letters

Words - simply and respectfully

The structure of the tray - in a system that attracts everyone

Color management - makes it attractive Colors

The benefits of the poster

1. Display the ideas of the poster.
2. These help students to interact with themselves.
3. Students engage in learning through each other's posters.
4. These help to improve critical transferable skills.

5.4.7. Newspapers

Newspaper is a rich source of information. It communicates authentic and first-hand information. Any information was immediately reported by the newspapers. Photographs, recent scientific information, new strategies are all published by the newspapers. Newspapers consist of the following news.

1. General news
2. Political news
3. Education news
4. Area news

5. World news
6. Sports news
7. Weather news
8. Job news
9. Advertisement news
10. Breaking news

Contents of the newspaper items relates to the school curriculum. So the teacher can use the news in classroom and may use a bulletin board to display the newspaper cuttings on a particular topic. The display news could be changed every day. Students can be given the task of writing daily importance news items on the chalkboards in the corridor.

Student can prepare files of newspaper cuttings on different Computer science topics or issues. These cuttings can then be used for conducting group discussions, giving group assignments or even for individual study. These files come handy when you require information on a particular aspect of a topic. The files prepared by the teacher and students can be maintained in the school library.

Benefits

- Newspapers are an adult medium and students all ability levels are inherently proud to be seen reading them.
- Newspapers deal with contemporary issues which are happening here and now and hence provide motivation for reading and discussion.

- Newspapers make learning 'fun'.
- Newspapers are extremely flexible and adaptable to all curriculum areas and grade levels.
- Newspapers bridge the gap between the classroom and the "real" world.
- Newspapers build good reading habits that will last a lifetime.
- Newspapers can be cut, marked, clipped, pasted, filed and recycled.
- Newspapers give everyone something to read – news, sports, weather, editorial and comics.
- In the long run, newspapers are cost effective and provide education.
- Newspapers contain practical vocabulary and are the best models of clear, concise writing.

5.5 Use of Mass Media for Classroom instruction

- 1. Newspapers**
- 2. Magazines**
- 3. Radio**
- 4. Television**
- 5. Internet**

1. Newspapers

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2. Magazines

A magazine is something that lies between a newspaper and a book. Unlike the book it is published periodically in a series. But it differs from a newspaper. The reporting of day-to-day happenings is not its main

concern. To inform, instruct and entertain through comments on chosen news and features on subjects of interest is what it is engaged in.

The comments are not cursory and hastily put down as in a newspaper as there is relatively longer time lag between the story happening and publishing. The time lag gives ample time for putting together well thought of and synthesized reactions. Newspapers are read on the day and thrown away the next day. Magazines are read and reread and passed around to others or stored away for future reference. They travel far and reach a wider audience. There are many types of magazines. They are as follows.

- New Magazines
- Sports Magazines
- Cultural Magazines
- Intellectual Magazines
- Magazines for children
- Magazines for men
- Magazines for women
- Business Magazines
- Educational Magazines

Educational Magazines

- It provides a variety of reading materials which exposes children to literary genres including poetry,

photo stories, short stories, fiction, non-fiction and current event articles.

- It supplements classroom instruction by providing additional material in different subjects.
- It includes writing activities that reinforce reading skills and improve grammar, exercises that develop spelling, vocabulary and word analysis skills, puzzles and riddles that develop critical thinking and reasoning power and activities for numerical data processing and chat-reading skills.
- It leads to professional development of teachers in classroom management techniques, subject knowledge and instructional practices and strategies.
- It provides opportunities for independent student reading practice and at home reading assignments.
- It provides a means of proper utilization of leisure time for students and act as source of entertainment.
- It enriches the learner with news about the outside world and heritage and culture.
- It can foster moral and spiritual values and develop national and international understanding.

3. Radio

Radio is an important teaching aid for Computer Science. Though Radio is used for the purpose of entertainment, these days it is also used for teaching

purposes. A comprehensive course in language can also be presented through Radio.

Radio technology offers a unique way for teachers to integrate technology into the curriculum. Teachers without Television or internet connections will find radio an accessible technology for bringing the world to their students. It brings the outside world into the classroom making educational programmes more attractive and entertaining. Now-a-days, Content-based computer science topics are available in the radio channel of All India Radio, BBC and the teacher has to make arrangements for the students to listen to the programmes. If necessary the teacher may record the programme in a tape recorder and replay it on some other occasion.

Uses

- A radio is very useful to the language teacher because, it bring in experts in language teaching whose instructions will be quite helpful to the students.
- All kinds of listeners at all places can use the radio profitably.
- All India Radio broadcasts Educational programmes and they also give the topic, date and time in advance, which is very helpful.
- It is very easy to accessible and has a wide coverage.
- The capital investment of it is very low and even the operating cost is also not so high.

- Listening to English news and other programmes helps the students to improve their pronunciation.
- With the help of radio, the students listening and speaking comprehension can be improved.
- Lectures, talks and addresses of important personalities from any corner of the world can be heard on Radio.
- Children general knowledge widened.

4. Television

Television is the most powerful medium of communication. It has revolutionized the method of teaching and learning. It is a convenient and economical medium of reaching a large cross-section of population. It combines the best of radio and motion pictures. It helps overcome barriers to learning. There are various programmes which provide up-to-date news around the world like Discovery Channel, Animal planet, BBC, CNN, National Geographic, ESPN, Ten Sports, Pudiya thalamurai, Doordarshan and so on.

Now-a-days, it has become the most recent audio-visual aid for classroom instruction. This is a multi-sensory audio visual aid, which can be effectively used for language teaching. The Directorate of School Education, Government of Tamilnadu, Chennai and the Doordarshan Kendra, Chennai has been telecasting programmes for teachers and students.

There are many educational television programmes in India like:

- a. Secondary school television project (STV)
- b. Delhi Agriculture Television Project (Krishi Darshan) (DATV)
- c. Satellite Instructional Television Experiment (SITE)
- d. Post – SITE Project
- e. Indian National Satellite (INSAT)
- f. Higher Educational Television Project (HETV) of UGC

Educational Television

The television is called as the “queen of audiovisual aids”. It is a one of the mass communication. It appeals to both the eye and the ear. There are three types of educational television programmes.

1. Direct teaching enrichment programmes
2. Supplementary enrichment programmes
3. Demonstration type programmes

There is a recent development in Educational television programme. They are:

- Satellite based communication
- Telephone clubbed with television
- Video tapes and
- Multimedia packages

Uses

- A large number of students can be given information at a time.
- It can help supplement class room oral teaching.
- It develops in them positive and self-confidence towards the foreign language by listening and seeing the English channels.
- Television can be used for telling stories, teaching of structures and sentences, their forms and patterns.
- It also inculcates and develops a sense of critical appreciation among the young learners by showing correct and modulated poetry reading.
- Television programme can highlight the function of the speech organs in producing right speech sounds.
- Listening, speaking and understanding abilities of the learners can be improved.
- The students who are not able to attend the class can watch and listen from their homes itself.
- It is time saving. Most of the syllabus may be covered in less time.
- Doordarshan programmes like UGC, Kanbom karpom are very helpful for students who are appearing for their public exams as the lessons and the question models are explained in detail.

Disadvantages

- It is a one-way communication only.
- Slow learners cannot cope up with it.

- Again there is a dearth for teaching who can explain the students before the programme commences.
- Poor accessibility and it is difficult to integrate the students.
- Sometimes visual becomes a distraction in the learning process.

5. Internet

Internet is the most user-friendly graphical system that offers huge amount of information to the users. Internet reveals huge sets of pages containing data, information, images, symbols, words, sounds, video clips etc., collected from all over the world. Internet is a network of networks. It is a group of two or more networks that are

- Interconnected physically
- Capable of communicating and sharing data with each other and
- Able to act together as a single network.

We can exchange text, data files, messages and programs with any other user.

Significance of internet

1. Procuring information on the internet

The information about people, cultures, societies, languages, nations, products, organizations, inventions, discoveries, games, events, research

data and findings, health and nutrition, developmental activities, media, politics, day-to-day news etc. is available on the internet.

2. Providing information on the internet

Individuals and institutions would like to re-present their ideas, praises, activities, programs, new-proposals, mission and vision statements, services offered, interests etc. over the internet. They can set up their home pages in a decorative and attractive way to extend their market and services. For this purpose, one can take up publishing, extension, teaching etc., on the net.

3. Compiling information on the internet

Internet can be used for obtaining specialized information from the web. One can generate opinions of people, experts, specialists, politicians, religious leaders, etc. by administering various instruments through e-mail or involving in group discussion, list servers, chatting groups, job groups, medical experts groups, etc.

Internet in Education

The internet has very wide applications in the field of education. It provides a number of learning experiences and educational resources to students, teachers, administrators, policy makers and all those who work for the cause of education. Some of the important ways in which the internet helps children are

- It helps develop and improve reading, writing, research and language skills.
- It provides support for kids with special needs and their parents.
- It is an exciting outlet for artistic expression.
- It provides information about every topic imaginable.
- It is a great way to hang out with your friends and be at home at the same time.
- It provides personal contact with new people and cultures.

The open learning methodology means the learners need not to go to the educational institutions. The internet provides a huge amount of information to the students across the world. So the internet can be used for delivering instructional inputs or providing information to the open school learners.

Internet will provide information to the schools and students in the following areas:

- Access to electronic information. Archives and databases.
- Direct support to teachers and learners to enhance educational access, quality.
- Career counselling
- Job requirements and opportunities available.
- Courses available through distance education and open learning mode.
- Provision of internet services.

- Provision of directory services.
- Offering courses through internet.

5.6 New Emerging Media

With the emergence of online technology, the production, distribution and consumption process of news has been changed. Newspapers and television channels have shifted their importance to the online editions. Each and every media houses in India now have their own websites. Especially the electronic communication with the internet driven technology has brought lots of innovations. The new forms of media have created opportunity for interactions between the producers and consumers of news. These new forms of media are known as New emerging media.

5.6.1 Tele-conferencing

Tele-conferencing is essentially a alive interactive audio, video or audio-video meeting that ensures between two or more participants who are at different places. Here participants communicate via telecommunicating networks using their tablets, mobile phones, laptops, computers and specially designed tech-enabled meeting rooms.

There are different types of teleconferencing. They are

1. Audio Teleconference
2. Audio Graphics Teleconference

3. Computer Teleconference
4. Audio - Video Teleconference

Let us see them briefly

1. Audio Teleconference

This is also called conference calling. Participants links via telephone lines. Meeting can be conducted via audio conference.

Distance learning can be conducted by audio conference.

2. Audio Graphics Teleconference

This telecommunication channels transit visual information such as graphics, documents and video pictures as an help to voice communication.

This type of communication is useful for meetings and distance learning.

3. Computer Teleconference

Here telephone lines are connected with two or more computers. Anything that can be done on a computer can be sent over the lines. Using electronic mail (Email) memos, reports, letters, lessons can be sent to anyone on the local area network (LAN) or wide area Network (WAN). Items generated by computer which are normally in printed form are sent by E-mail.

Computer conferencing is an emerging area for distance education. Students receive texts and work books via E-mail. Students download these files and use for further study.

4. Audio - Video Teleconference

Voice communication and Video images are used for conferences. It can display anything that can be captured by a TV camera. Here you can see and hear person on a television monitor. Participants face-to-face meeting enables to see the facial expressions and physical behaviour of participants at remote sites. Graphics are used to enhance understanding.

Video conferencing is an effective way to use one teacher to teach to a number of sites. Video conferencing maximizes efficiency because it provides a way to meet the several groups in different locations, at the same time.

Tools needed for Video Teleconferencing

To get started with Video Teleconferencing and connect your sites, you need the following few things.

- A desktop computer, laptop or smart phone
- An internet connection
- A microphone and webcam
- An external speaker
- Video Telecommunication software

Advantages of Teleconferencing

1. Saves Time :

The most important advantages of Teleconferencing is time-saving. It is possible to hold the meeting on a very short notice. In addition there is

no time spent to travel to the venue of the meeting as all meetings are held through internet and computers.

2. Save on travel expenses :

Now-a-days travel costs can be quite significant. Teleconferencing removes the need to travel and saves huge money.

3. Efficient Record Keeping :

One of the main advantage of Teleconferencing is efficient meeting Record keeping. Computer devices are able to record, keep every details of the meeting. Teleconferencing also makes it very easy to retrieve this data in the future and make references whenever necessary.

4. Cut conference costs :

There are many costs to hold a meeting. Hiring venue, buying food and refreshments etc. associated with any meeting. Teleconferencing removes all these costs as no such arrangements need to be made.

5. Reliability :

Teleconferencing is one of the most reliable ways of holding meeting. Due to advancement in technology, teleconferencing channels are today much more stable. It is also very secure mode of communication where the safety of data as well as privacy is guaranteed.

6. Services :

Health care management, Business, Education, International conference and on line interviews are held using Teleconferencing.

5.6.2 Communication Satellites

Communication satellite is an artificial satellite that relays and amplifies radio communication signals via transponder; it creates a communication channel between a source transmitter and a receiver at different locations on earth. Communication satellites are used for television, telephone, radio, internet, Metrology and Military applications all over the world. As on 1st January 2021, there are 2224 Communication satellites in Earth orbit. Most communication satellites are in geostationary orbit 35,900 kms above the equator so that the satellite appears stationary at the same point in the sky.

Communication satellites use a wide range of radio and microwave frequencies. To avoid signal interference, international organisations, have regulations for which frequency ranges or 'bands' certain organisations are allowed to use. This allocation of bands minimizes the risk of signal interference. These satellite provide communication links between various points on Earth. Watching your TV shows or movies from internet would have been impossible without this.

Satellites and Education

India launched Education Satellite GSAT-3 known as EDUSAT on 20th September 2004 from Sriharikota, India. It is meant for distant classroom education from school level to higher education. This was the first dedicated Educational satellite that provide the country

with satellite based two way communication to classroom for delivering educational materials.

Objectives of EDUSAT

- Providing support to formal and non-formal education.
- Teachers' training program
- Increasing access to quality resource persons
- Enhancing community participation
- Taking education to remotest corner of the country.

Application of EDUSAT

- Conventional Radio and Television broadcasting
- Interactive Radio and Television
- Exchanging of Data
- Teleconferencing both one way or two way & Audio conferencing.
- Computer conferencing
- Web based education/

EDUSAT Worlds' first satellite meant only for educational purposes is being used. Training by teleconferencing overcomes the problem of transmission loss since participants are watching, listening and interacting with primary resource persons. CIET a Constitute unit of NCERT using this technology of teleconferencing over a decades to give training / orientation to teachers and teacher educators. Other institutions like IGNOU, Vigyan Prasar, Sarva Shiksha

Abhiyan (SSA), Rehabilitation Council of India (RCI) etc are using EDUSAT.

Advantages of Communication Satellites

1. Flexibility
2. Ease in putting in new circuits
3. Distances are effortlessly taken care of and expense does not make a difference.
4. Broadcasting Conceivable Outcomes
5. Each and each side of the earth is secured.
6. User can control the system.

Disadvantages of Communication Satellites

1. The introductory costs are excessively high
2. Congestion of frequencies
3. Interference and proliferation

5.6.3 Computer Networking

A computer network is a set of computers sharing resources located on or provided by network nodes. The computers use common communication protocols over digital interconnections to communicate with each other. These interconnections are made up of telecommunication network technologies, based on physically wires, optical, and wireless radio-frequency methods that may be arranged in a variety of network topologies.

The nodes of a computer network may include personal computers, servers, networking addresses,

and may have host names, Host names serve as memorable labels for the nodes, rarely changed after initial assignment. Network addresses serve for locating and identifying the nodes by communication protocols such as the internet protocol.

Computer networks may be classified by many criteria, including the transmission medium used to carry signals, bandwidth, communications, protocols to organize network traffic, the network size, the topology, traffic control mechanism and organizational intent.

Computer networks support many applications and services such as access to the world wide web, digital video, digital audio, shared use of application and storage servers, printers and fax machines and use of email and instant messaging applications.

A computer networks extends inter personal communications by electronic means with various technologies such as email, instant messaging, online chat, voice and video telephone calls and video conferencing. A network allows sharing of network and computing resources. Users may access and use resources provided by devices on the network such as printing a document on a shared network printer or use of a shared storage device. A network allows sharing of files, data and other types of information giving authorized users the ability to access information stores on other computers on the network. Distributed computing uses computer resources across a network to accomplish tasks.

Types of Computer Network

A computer network is mainly of four types. They are

1. LAN (Local Area Network)
2. PAN (Personal Area Network)
3. MAN (Metropolitan Area Network) and
4. WAN (Wide Area Network)

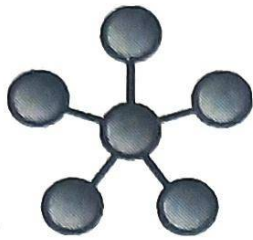
Need of computer networking

Computer networks help users on the network to share the resources and in communication. Networking of computers help the network users to share data files. Users can share devices such as printers, scanners, CD-Rom drives, hard drives etc.

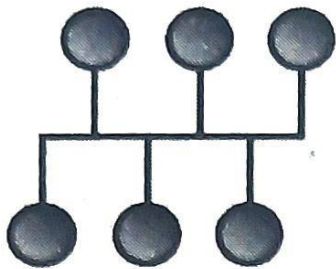
Types of network Topology

Topology is derived from two greek words topo and logy where topo means place and logy means study. In computer network topology is used to explain how a network is physically connected. A Topology mainly describes how devices are connected with internet and other communication links. Network topology defines the layout, virtual shape or structure of the network. Types of network topology are

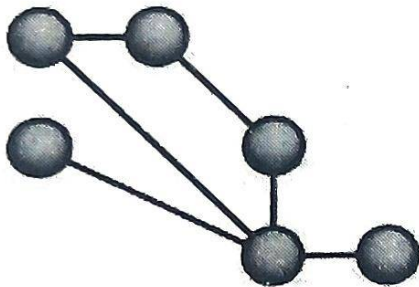
1. Star Network



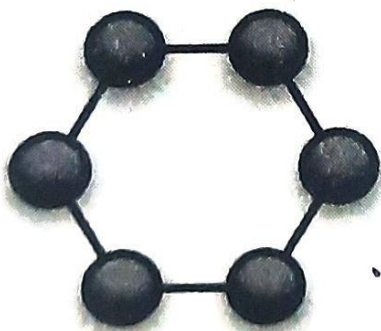
2. Bus Network



3. Mesh Networking



4. Ring Network



Advantages of Networking

1. Strengthen business connections
2. Get fresh ideas
3. Raise your profile
4. Get access to job opportunities
5. Gain more knowledge
6. Get Career advice and support
7. Built confidence
8. It boosts storage capacity of the computer
9. It makes file sharing easier
10. It is an inexpensive system

Disadvantages of Computer Networking

1. The connected systems on a network entirely depend on the main server. If there is any problem in server, complete network failed.
2. Since networks are based on the centralized server, most of the decisions are made themselves.
3. Setting up Computer network can be expensive
4. Due to interactions, a virus can easily spread between computers in a network.
5. In networking, large number of users are using computers. Hackers can easily access the network using specialized tools.
6. It can drastically decrease the productivity of the company.

5.6.4 Word Processors

Word processor is a computer program used to write and revise documents, compose the layout of the text and preview on a computer monitor how the printed copy will appear. The last capability is known as "what you see is what you get".

Word processors facilitate writing and editing, especially with their ability to copy and move text, (Cut and paste) their built-in dictionaries to check spelling, and their grammar checkers. Other common features include a wide choice of typographic fonts and sizes, various paragraph and page layouts, tools for finding and replacing strings of characters and word counts. Modern word processors also have many features one reserved for desktop publishing systems, such as table creation and importation of graphic images. They typically provide templates for common document types, such as letters, memos and resumes and can generate multiple copies of a document with recipient addresses drawn from a list. (Mail merge)

Desktop publishing programs have word processing features but also provide highly flexible layout and control over appearance to combine text and graphics for advertising copy, magazines and books.

Examples of Word processor softwares

- Abiword
- Apple iwork-pages
- Apple Text Edit
- Coral word perfect

- Dropbox paper
- Google Docs
- Libe office → writer
- Microsoft office → Microsoft word

Functions of word processors

The functions of word processing described below are all basic ones if you have reasonably computer literature.

- Creating, editing, saving and printing documents.
- Pasting, moving and deleting text with in a document.
- Formatting text such as font type, size, bolding, underlining or italicizing.
- Creating and editing tables
- Adding symbols.
- Changing the text appearance
- Changing the page appearance

Advantages of Word processors

1. A document can be stored in a computer for a future reference.
2. Have special editing tools namely spelling and grammar checkers.
3. One can easily insert or replace a word or phrase without affecting the neatness of a document.

4. You can save your work and come back to it at a later time.
5. Plenty of professional quality document templates available online.
6. Can import a data from a database and use it to create mail merge.
7. Teacher and students gain a sense of security about losing assignments.
8. Teacher benefit by receiving a readable copy that is easy to grade for students.
9. Very helpful for publishing work
10. Work done on a word processor and saved on the internet is highly portable and accessible from anywhere.
11. Environment improves as there is no use of paper.
12. It helps to reconstruct complex office work.

Disadvantages of Word processors

1. Cost of word processor is very high.
2. Most users will probably never use 50% of the functionality of word.
3. Some functions are not always intuitive and it can take a while to get your desired effects.
4. You always need an internet connection to access your data files.
5. Security measures can be breached.

5.6.5 Blended learning

1. Introduction

Blended Learning is about effectively integrating ICTs into course design to enhance the teaching and learning experiences for student and teachers by enabling them to engage in ways that would not normally be available in their usual environment, whether it is primarily face-to-face. Blended Learning technologies are as follows:

- Broaden the space and opportunities available for learning.
- Support course management activities.
- Support the provision of information and resource to students.
- Engage and motivate students through interactivity and collaboration.

The learning and teaching activities need to be meaningful and relevant for the student learning.

Definition:

- The thoughtful integration of classroom face-to-face learning experiences with online learning experiences.
- It combines online with face-to-face learning. The goal of blended learning is to provide the most efficient and effective instruction experience by combining delivery modalities.

- Mixed mode or hybrid-learning is the integration of face-to-face (F2F) learning with online learning activities.
- A solution that combines several different delivery methods, such as collaboration software, web-based courses, EPSS and knowledge management practices.

2. Objectives

- Increase the amount and quality of faculty-to-student and student-to-student interaction.
- Increase opportunities for active and collaborative learning assessment before, during and after lectures.
- Help students prepare for class discussions or lab work.
- Facilitate more varied and engaging media for presenting course content.
- Address learning bottlenecks via new types of interactive and independent learning activities.
- Drive collaborative teacher problem-solving.

Uses :

- Blended learning can increase access and flexibility for learners, increase level of active learning and achieve better student experiences and outcomes. For teaching staff, blended learning can improve teaching and class management practices.

- **Enhanced learning:** Students usually receive more feedback and more frequent feedback, for their instructors. Students can acquire useful skills from using the Internet and computer technology (live instructor presence, technological enhancement). Students have more time to reflect and refer to relevant course and other research materials when working and writing online than when responding in class.
- **Support for student collaboration:** Real-time real-space interactivity and asynchronous online interactions, synchronous online interactions.
- **Course accessibility:** Students have access to unlimited up-to-date resources available via the web. Live signing in live courses, transcription of audio and video, labeling of images with alt text online.
- **Learner convenience:** Students have greater time flexibility, freedom and convenience by working part of the time online. Time flexibility without full-term scheduling, a synchronicity (Non-real-time learning), and the repeatability of some automated learning and ease of accessing course materials.
- **Rich learner assessment:** Automated assessments, live instructor-led assessments, peer assessments live or via the online classroom.
- **24/7 Accesses:** Students often develop or enhance skills in time management, critical thinking and problem solving. Students typically have 24/7 access to online course materials.

- **Multi-use/Dual use resources:** Course contents may often be used in both contexts (F2F and online)

5.6.6 Flipped Classroom

Flipped Classroom is an approach that allows teachers to implement a methodology or various methodologies in their classrooms. It means "School work at home and home work at school". Flipped classroom inverts traditional teaching methods, delivering instruction online outside of class and moving "homework" into the classroom.

Definition

Flipped Classroom is a pedagogical approach in which direct instruction moves from the group learning space to the individual learning space and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creativity in the subject matter.

It is called flipped because the traditional definition was "what used to be class work (lecture) is done outside the class (usually with videos at home) and what used be home work (assigned problems) is now done in the class".

The flipped classroom is a pedagogical approach in which typical lecture and home work element of a course are reversed. Short video lectures of instructional content are viewed by students at home before the class session, while in the class time is

developed to solve problems exercised, projects or discussions with the guidance of the teacher.

Frame work of flipped classroom

It is a student centred model in which class time explores topics in greater depth and create meaningful learning opportunities. In a flipped class room content delivery may take a variety of forms. Video lessons prepared by the teachers or others are used to deliver content although on line collaborative discussions and text readings may be used. More time can be spent in class on higher order thinking skills such as problem - finding, collaboration, design and problem solving. Students take different problems, working in groups construct knowledge with the help of their peers and teachers. Teachers interaction with students in a flipped classroom can be more personalised and less didactic.

Flipped classroom apply a mastery learning model that requires each student to master a topic before moving to the next one.

Four pillars of Flipped classroom

- F – Flexible Environment
- L – Learning Culture
- I – Intentional Content
- P – Professional Educators

1. Flexible Environment

The flipped classroom must allow for a variety of learning models. Educators will often physically rearrange their learning space to accommodate the lesson unit. Additionally flipped educators have to be comfortable with flexible learning environment in which the student choose what and where they learn outside of the classroom.

2. Learning Culture

In the flip learning model, educationists are more cognizant of the content and they choose to deliver and determine what direct instruction or traditional lecture can be shifted. Students are empowered to find content on their own, while educator help them make connection and build on their own understanding and proficiency. Parents and community must be kept engaged and informed for the culture to shift in this new model.

3. Intentional Content

Educators must continuously think about how they can use the flip learning models to help the student's gain conceptual understanding as well as procedural fluency when needed. They must evaluate what they must need to teach and what the materials students should explore on their own. Educators must adopt various methods of instruction such as active learning strategies, peer instructions, problem based learning or mastery or Socratic Method depending on grade level and "Subject Matter".

4. Professional Educators

The role of the teacher is even more important and it is often more demanding in flipped classroom than in a traditional one. During class time, teacher continually observe their students, providing them with feedback relevant in the movement and assessing their work. There is a deliberate shift from lecture centered to a student-centered class. Also educators must be able to connect with each other to improve their trade, to accept constructor criticism and to tolerate controlled classroom chaos.

Flipped classroom

The flipped classroom describes a reversal of the traditional teaching where students gained first exposure to new material outside of class, usually via reading or lecture, videos and then class time is used to do the harder work of assimilating that knowledge through certain strategies such as

1. Problem – solving
2. Discussion and
3. Debates

Advantages

1. Students take control of their learning.
2. Student can develop foundation of factual knowledge.

3. Students can understand facts and ideas in the context of a conceptual framework.
4. They can enrich their knowledge in ways that facilitate retrieval and application.
5. It saves time. Teacher may spend more time with students.
6. Students learn at varying speed.
7. Ability to rewind, save, review lesson plans at any moment.
8. Students are provided opportunities for review lesson front-load for classroom activities.
9. Patents can view lessons and better assists students.
10. Students do not struggle with completing homework because they forgot how it does.
11. Students are actively working with their peers.
12. Students take ownership of their learning.
13. Personalized educations for slow and fast learners.
14. Stronger bonds with teachers.
15. It provides adequate learning opportunities for verbal learners and for visual learners.

5.6.7 Artificial Intelligence

Artificial Intelligence (AI) broadly refer to any human-like behaviour displayed by a machine or system. In AI most basic form, computer are programmed to 'mimic' human behaviour using extensive data from past example of similar behaviour. With AI, machine can work efficiently and analyse vast amounts of data in the blink of an eye, solving problems through supervised, unsupervised or reinforced learning..

AI is now part of our daily lives. We have AI solutions for quality control, video analytic, speech-to-text and autonomous driving, as well as solutions in healthcare, manufacturing, financial services, entertainment and education.

History of Artificial Intelligence

Before 1949, computers could execute commands, but they could not remember what they did as they were not able to store these commands. In 1950, Alan Turing discussed how to build intelligent machines. Research done in this field from 1960 and laboratories had been established around the world. By 2000 solutions developed by AI researchers, were being widely used.

Application of AI

Artificial intelligence, the intelligence exhibited by machines has been used to develop thousands of applications to solve specific problems throughout industry and academic.

Here are some of the real-world applications of AI in Education.

1. Automation of basic administrative activities

A lot of time is spent by teachers in administrative activities like grading and assessment of worksheets. The use of AI in education can help automate the grading and assessment of activities like multiple choice questions, fill in the blanks etc. Another tedious and cumbersome activity for teachers is preparing the report cards of students. The use of AI in education can help automate this as well. Automation of administrative activities means teachers can spend more time with the students, thus making the learning process more efficient.

2. Personalized learning

The purpose behind application of AI in education is not to replace teachers, but give them a helping hand in understanding the potential and limitations of each student. The use of AI in schools make things easy and convenient for the teachers and students as well. By understanding the needs of every student, teachers can come up with a study plan for every students.

3. Constructive feedback

Programs powered by AI can provide valuable feedback both to the students as well as teachers. Use of AI in Classrooms can point out to teachers how to improve the instructions provided to the learners and also how to make learning more fun and interesting. Instant feedback to students helps them understand

where they are going wrong and how they can do it better.

4. Personalization

What better way to offer more personalized learning opportunities for students than to have. AI be able to analyze student responses, determine areas of need and interest, and find resources or create new questions to help students to greater understanding of the content.

5. Communication

Students and teachers will be able to communicate instantly with one another as well as to connect with other forms of AI around the world.

6. Accessible by all students

Due to socio economic status, transportation and disabilities may change how much access a student has to the classroom. However learning can happen anywhere with the help of AI.

7. Educational softwares are developed to student needs

AI gives importance to individualized learning. AI responds to the needs of the student, put greater emphasis on certain topics, repeating things that students have not mastered and generally helping students to work at their own pace.

8. AI may change the whole education system

AI has the potential to change about everything on education. Students can learn from anywhere in the world to anytime and sometimes AI may replace teachers in some instances. Educational programs powered by AI are helping to learn basic skills and they will likely to offer students a much wider range of services.

Advantages of Artificial Intelligence in Education

1. AI-based education can provide an opportunity to study at any time.
2. AI-based solutions can adopt according to the level of knowledge, interesting topics etc. The system tends to help students with their weak side.
3. AI based education offer virtual mentors to track the students' progress.
4. AI-enabled education offers appropriate teachers, depending on the teaching experience and soft skills.
5. AI operates 24 x 7 without interruption or breaks
6. AI augments the capabilities of differently abled individuals.
7. It introduces a new technique to solve new problems.
8. It handles the information, better than humans.
9. It improves work efficiency.
10. It uses more powerful and more useful computers.

Disadvantages of Artificial Intelligence in Education

1. The implementation cost of AI is very high.
2. AI based teaching lead to serve unemployment.
3. Development of software for AI is slow and expensive.
4. It lacks creativity and decreases the thinking power of the students.
5. Human interaction decreases.

5.6.8 Augmented Reality

Augmented Reality is an interactive experience of a real world environment where the objects that reside in the real world are enhanced by computer generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory or other sensory information.

Augmented Reality is a technology in which a computer-generated image is super imposed on to the user's vision of the real world, giving the user additional information generated from the computer model. Using an AR system, the user's view of the real world is enhanced.

The basic idea of augmented reality is to super impose graphics, audio and other sensory enhancements over a real world environment in real time. Some of the most exciting augmented reality work began taking place in research labs at universities around the world. In February 2009, Mistry presented

Via this learning, the students are able to make the most of their understanding skills and imagination skills make them be interested in the subject where they get inclined towards the concept's deep learning instead of rote learning.

Advantages of Augmented Reality

1. It reduces line between real world and virtual world.
2. It enhances perceptions and interactions with the real world.
3. Due to its use in medical industry, life of patients have become safer. It helps in efficient diagnosis of diseases and in early detection of them.
4. It can save money by testing critical situations in order to confirm their success without actually implementing in real time. Once it is proven, it can be implemented in real world.
5. It can be used by military people without putting their life in danger by way of battle fields simulation before the actual war. This will also help them in actual war to take critical decision.

Disadvantages of Augmented Reality

1. It is expensive to develop the AR technology based projects and to maintain it. More over production of AR based devices is costly.

2. Lack of privacy is a concern in AR based application.
3. In AR, people are missing out an important moments.
4. Low performance level is a concern which needs to be addressed during testing process.
5. It requires basic learning to effectively use AR compliant devices.

Review Question

1. How print resources can be used for teaching History?
2. How Mass Media can be used for teaching History.
3. What is Multi-sensory Teaching? Explain its various forms.
4. What is Tele-conferencing? Write its merits and demerits.
5. What is Artificial Intelligence? How it is used in Teaching.
6. Explain in detail Augmented Reality and how it is used in Education?
7. Write notes on Communication Satellites? How it can be used for instruction?
8. Explain Edgar Dale's Cone of Experience.

**B.Ed. DEGREE (2 YEAR) EXAMINATION,
February 2022**

**(For the candidates admitted during the Academic
Year 2021-2022 only)
First Year - First Semester**

PEDAGOGY OF HISTORY: PART - 1

Time : 3 hours

Maximum marks : 70

**PART A - (5 × 1 = 5 marks)
Answer ALL questions.**

I. Objective Type Questions.

1. வரலாறு என்ற சொல் எந்த மொழியிலிருந்து வந்தது ?

(அ) கிரேக்கம்

(ஆ) இலத்தீன்

(இ) சீனா

(ஈ) சமஸ்கிருதம்

Which language the word history comes from?

(a) Greek

(b) Latin

(c) China

(d) Sanskrit

2. Micro-Teaching என்ற வார்த்தையை கண்டுபிடித்தவர் யார் ?

(அ) பிளாட்டோ

(ஆ) ஜான் டூயி

(இ) டிவைட் N. ஆலன்

(ஈ) இம்மானுவேல் கான்ட்

Who was the inventor of the word Micro-Teaching?

(a) Plato

(b) John Dewey

(c) Dwight N. Allen

(d) Immanuel Kant

3. மாரிசன் கற்பித்தல் மாதிரியில் எத்தனை படிநிலைகள் உள்ளன ?

(அ) 4

4

(ஆ) 5

5

(இ) 3

(ஈ) 8

How many steps are there in Morrison Teaching Model?

- (a) 4 (b) 5 (c) 3 (d) 8

4. செயல்வழிக் கற்றல் (ABL) முறையை சென்னையில் அறிமுகப்படுத்தப்பட்ட ஆண்டு _____.

- (அ) 2003 (ஆ) 2005 (இ) 2001 (ஈ) 2007

ABL (Activity Based Learning) method was introduced in Chennai in the year of _____.

- (a) 2003 (b) 2005 (c) 2001 (d) 2007

5. எட்கர் டேல் என்பவர் _____ நாட்டைச் சேர்ந்த கல்வியாளர்.

- (அ) அமெரிக்கா (ஆ) பிரான்சு
(இ) ஜெர்மனி (ஈ) கனடா

Edgar Dale was _____ Educator.

- (a) An American (b) French
(c) German (d) Canadian

PART B - (3 × 5 = 15 marks)

Answer any THREE questions.

(Maximum 250 words or two and half a page for each question)

6. உயர்நிலைப் பள்ளியில் வரலாறு கற்பித்தலின் நோக்கங்கள் யாவை?

What are the aims of teaching history at high school level?

7. வினா கேட்கும் திறன் பற்றி எழுதுக.

Write a notes on the skill of Questioning.

8. காலவரிசை அணுகுமுறை என்பது என்ன?

What is chronological approach?

9. செயல்வழிக் கற்றல் என்றால் என்ன ?

What is activity based learning?

10. எட்கர்டேலின் முக்கோண அமைப்பைப் பற்றி வரைக.

Sketch out the Edgar Dale's cone of Experiences.

PART C - (5 × 10 = 50 marks)

Answer ALL questions.

(Maximum 500 words or 5 pages for each question)

11. (அ) கல்வி நோக்கங்களில் புளுமின் வகைப்பாடுப் பற்றி விவரி.

Explain the Bloom's taxonomy of Educational Objectives.

Or

(ஆ) வரலாறு கற்பிப்பதால் ஏற்படும் பயன்களைப் பற்றி ஓர் கட்டுரை வரைக.

Write an essay on values of Teaching History.

12. (அ) நுண்ணிலைக் கற்பித்தலில் உள்ள ஏதேனும் எட்டுத் திறன்களைப் பற்றி விவரி.

Describe any eight micro-teaching skills.

Or

(ஆ) நுண்நிலைக் கற்பித்தலின் கருத்து, படிநிலைகள், கற்றுச் சொல் முறைகள் மற்றும் பாடத்திட்டம் ஆகியவற்றைப் பற்றி எழுதுக.

Write an concept, steps, cycle and lesson plan of Micro-Teaching.

13. (அ) ஒன்பதாம் வகுப்பு சமூக அறிவியல் பாடத்தில் உள்ள ஏதேனும் ஒரு அலகிலிருந்து வரலாற்றுப் பாடத்திட்டத்தை எழுதுக.

Write a lesson plan from any unit in IX standard subject.

Or

(அ) வரலாற்றுப் பாடத்தில் உள்ள பொதுமைய முறை மற்றும் அலகுமுறையில் உள்ள நிறைகள் மற்றும் குறைகளை விவரி.

Explain the merits and demerits of concentric approach and unit approach in History subject.

14. (அ) வரலாறு கற்பித்தலில் உள்ள விரிவுரை முறை மற்றும் புதிர்தீர்முறையினை விளக்குக.

Explain the lecture method and problem solving method in teaching history.

Or

(ஆ) வரலாறு கற்பித்தலில் உள்ள கலந்துரையாடல் முறை மற்றும் அலகுமுறையில் உள்ள நிலை, குறைகளை எழுதுக.

Explain the discussion method and unit method in Teaching History.

15. (அ) வரலாறு கற்பித்தலில் வன்பொருள், செயல்முறை துணைக் கருவிகளின் பங்களிப்பினை விவரி.

Discuss the hardware instructional aids in teaching history.

Or

(ஆ) வரலாறு கற்பித்தலில் மென்பொருள் செயல்முறை துணை கருவிகளைப் பற்றி விவரி.

Discuss the software instruction aids in teaching history.

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